

COMPACT SUPPORT OF AN ISENTROPIC IMPLOSION THE CASE $\gamma \geq 2$

ABSTRACT. A compactly supported physical-vacuum realisation of a radial isentropic self-similar implosion is constructed for adiabatic indices $\gamma \geq 2$. Coincidence with the profile on the sonic cone is proved by Kato theory. Existence of the vacuum collar uses the model-energy embeddings of Coutand–Shkoller, *Comm. Pure Appl. Math.* 64 (2011), Lemma 3.1 and the $\gamma = 2$ energy of their (1.13), transferred to a chart with one lock and one vacuum end. Theorem 8.1 is not used. This is not a resolution of the Clay Navier–Stokes problem.

1. SETTING

Radial isentropic Euler, $d \in \{2, 3\}$, $\gamma \geq 2$. Profile: Merle–Raphaël–Rodnianski–Szeftel [4] or Buckmaster–Cao–Labora–Gómez-Serrano [1], singular only at the origin at time zero. Sonic ray $r_s(t) = Z_s(-t)^\lambda$. Lock $\sigma(t) = \bar{J}_+(t, \bar{\eta}(t, \bar{a}_*))$.

2. BRIDGE

Z_0 first, then L . $r_{\text{in}} = Z_0(-t_0)^\lambda$ forces $\max_{k=1,2,3} |\partial_r^k(u, c)|_{r_{\text{in}}} < \text{Lip}_*/8$. Then $L > (35/16)|q_0^{\text{br}} - q_0^{\text{in}}|/(\text{Lip}_*/2)$. $D_t J_+ = -c \partial_r J_+ - c \frac{d-1}{r} u$, $\det = c^6 > 0$. Degree-7 Hermite, $C_* = 35/16$.

$$\text{Lip}_* = \min(1/|t_0|, \mu(t_0)(-t_0)^{-\lambda}/4).$$

3. COLLAR

$V_* = \{\varphi \in H^1(I) : \varphi(a_*) = 0\}$. Edge free: $v(1) = \dot{R}$. Cutoff $\omega \equiv 0$ near the lock, $\omega \equiv 1$ near vacuum. Model energy $\mathcal{E} = \mathcal{E}_{\text{in}} + \mathcal{E}_{\text{vac}}$ ($a_0 = 0$). No fractional distance weights.

4. ENERGY

Fluxes die at a_* by $\varphi(a_*) = 0$ and at $y = 1$ by $p = \rho_0 = 0$. On $\{\omega = 1\}$, $\partial_y p / \rho_0 = \partial_y h_0 \in L^\infty$. Row 4 of $\mathcal{C}_4$ is then a model-energy pairing. Convective Leibniz

$$[w \partial_y, \partial_t^4]w = - \sum_{k=1}^4 \binom{4}{k} (\partial_t^k w) (\partial_t^{4-k} w_y)$$

is $C(M_0)\mathcal{E}$ on the interior chart by $H^2 \hookrightarrow C^1$, and $C(M_0)\mathcal{E}$ on the vacuum sliver by the one-sided $\gamma = 2$ embeddings of [2] (energy (1.13) and Lemma 3.1), which give $\|w_y\|_\infty \leq C(M_0)$ in one dimension. Those embeddings do not use a second vacuum end. Interface on I_ω : $\rho_0 \geq \rho_\dagger > 0$, error $O(\kappa)\mathcal{E}_{\text{in}}$.

$$\begin{aligned} (1) \quad \frac{d}{dt}(\mathcal{E}_{\text{in}} + \mathcal{E}_{\text{vac}}) + \frac{7}{8} \kappa \int_I \omega \rho_0^2 |\partial_t^4 w_y|^2 \\ \leq C_1 \mathcal{E} + C \|\sigma\|_{W^{4,\infty}}^2, \end{aligned}$$

with $C_1 = C_{\text{in}} + C_{\text{vac}} + C_{\text{int}} < \infty$. Dissipation draw $\frac{1}{32} + \frac{1}{16} = \frac{3}{32} < \frac{1}{8}$.

$$T^* = \min\left(|t_0|, C_1^{-1} \ln\left(1 + \frac{C_1 M_0}{C_1 M_0 + C \|\sigma\|_{W^{4,\infty}}^2}\right)\right).$$

5. CONTINUATION

Mesh $t_k = t_0 + kT^*$ reaches $t = 0^-$ in finitely many steps: the collar stays in $\{r \geq r_{\text{in}} > 0\}$ and $\|\sigma\|_{W^{4,\infty}([t_0,0])} < \infty$. $N \rightarrow \infty$, $\kappa \rightarrow 0$: Aubin–Lions on $\{\omega = 1\}$, Rellich on $\{\omega = 0\} \cup I_\omega$. Lock exact on V_* , so the limit lies in $\mathcal{S}$. Cone $\mathcal{K} = \{(t, r) : r \leq Z_s(-t)^\lambda\}$. Friedrichs system in $U = (\frac{2}{\gamma-1}c, u)$, $A_0 = I$ [3]. On the profile, $\dot{r}_s = \bar{u} - \bar{c}$, so $\delta U = 0$ on $\mathcal{K}$.

Theorem 1. *Let $\gamma \geq 2$. Assume the one-sided model-energy embeddings of [2] as above, not Theorem 8.1. There is a compactly supported physical-vacuum solution on $[t_0, 0)$ with $(\rho, u) = (\bar{\rho}, \bar{u})$ on $\{Z \leq Z_s\}$, $\rho = 0$ for $r \geq R(t)$, and $R(t) > r_s(t)$.*

Remark 1. Not Clay. Not $\gamma \in (1, 2)$.

REFERENCES

1. Tristan Buckmaster, Gonzalo Cao-Labora, and Javier Gómez-Serrano, *Smooth imploding solutions for 3D compressible fluids*, Forum Math. Pi **13** (2025), e6.
2. Daniel Coutand and Steve Shkoller, *Well-posedness in smooth function spaces for the moving-boundary 1-D compressible Euler equations in physical vacuum*, Comm. Pure Appl. Math. **64** (2011), no. 3, 328–366, Lemma 3.1 and energy (1.13) for the $\gamma=2$ letter; Theorem 8.1 only for the γ in (1,2) blueprint; not Theorem 1.1.
3. Tosio Kato, *Quasi-linear equations of evolution, with applications to partial differential equations*, Spectral theory and differential equations, Lecture Notes in Math., vol. 448, Springer, Berlin, 1975, pp. 25–70.
4. Frank Merle, Pierre Raphaël, Igor Rodnianski, and Jeremie Szeftel, *On the implosion of a compressible fluid I: Smooth self-similar inviscid profiles*, Ann. of Math. **196** (2022), no. 2, 567–778.